# Kwan Hang (Gavin) Yam

legendgavin00@gmail.com · Los Angeles, CA · linkedin.com/in/gavin-yam2023 · Portfolio

Simulation engineer (UC Berkeley MEng, simulation and control) who builds and validates automated CFD, CHT, and FEA pipelines on OpenFOAM and CalculiX and presents them to customer CFD teams; prior hands-on ANSYS Fluent, HyperMesh, and NX Nastran.

## EDUCATION

**University of California, Berkeley** — MEng, Mechanical Engineering (Simulation and Control) *Aug 2023 – May 2024*

**California State Polytechnic University, Pomona** — B.S., Aerospace Engineering *Aug 2018 – May 2023*

## SKILLS

**Simulation:** OpenFOAM (snappyHexMesh), CalculiX, Gmsh, ANSYS Fluent, HyperMesh, NX Nastran, FEMAP, Simulink, multi-body dynamics
**Programming:** Python, MATLAB, C++, Lua; AWS (ECS, Lambda, CloudWatch), PostgreSQL
**Controls, embedded, and CAD:** Kalman filtering, ROS2, Arduino, BLDC motors, PCB design; SolidWorks, NX, Fusion 360, P&ID

## EXPERIENCE

**Khorium Inc** — Simulation Engineer *Los Angeles, CA · Feb 2026 – Present*

- Built Python orchestration for OpenFOAM and CalculiX across 7 solver templates (external/internal flow, URANS, CHT, thermal, structural); ran 100+ validation studies in 6 weeks through an agentic MCP interface
- Scored the full CAD-to-solve pipeline against public benchmarks: backward-facing step reattachment within 13% of Driver–Seegmiller, NASA wall-mounted hump separation within 1.3% of experiment; cross-validated against ANSYS Fluent and quantified a 1.24× wall-function friction bias at y+ ≈ 18
- Presented the meshing pipeline to a customer CFD team and handled live technical Q&A on rotating-region fan domains, AI auto-repair and geometry-change traceability, and mesh quality (4–8M cells in 5–10 min); the customer called the session a success and moved to hands-on testing
- Supported six customer and prospect engagements spanning EV battery, industrial CFD, appliance, and medical-device FEA: verification criteria, license and on-prem deployment constraints, solver selection, requirements gap analyses, and EU data-residency compliance
- Integrated Gmsh and snappyHexMesh boundary-layer meshing (coverage 1.45% → 100%, friction Cd 0.42 → 0.018) and diagnosed 12 platform defects from validation data, including a $10^6$× reference-area unit error and a silent 327× mesh-volume error that checkMesh passed

**Orbital Exploration Systems (ORBES)** — Mechanical Controls Engineer *Los Angeles, CA · Jun 2025 – Jan 2026*

- Designed linear control laws for a free-flying 6-DOF zero-G robot (reaction wheels and fans), reaching ±5° accuracy in compound maneuvers
- Implemented Kalman-filter visual-inertial odometry in ROS2 with hardware acceleration (+400% throughput); multiprocessing redesign saved $2K in embedded hardware and power

**ACIES Engineering** — Mechanical Design Engineer *San Jose, CA · Aug 2024 – May 2025*

- Designed heat-pump and VRF systems for 50+ commercial spaces (10% energy savings, CBC compliant) and walked clients through them on 25+ site surveys

## PROJECTS

**Honeywell International (contract)** — Senior Capstone Co-lead, Satellite Jitter Mitigation *UC Berkeley · Aug 2023 – May 2024*

- Modeled the point-acquire-track loop as a multi-body dynamics system in Simulink from 10 identified jitter sources; delivered passive and active mitigation to arc-second accuracy at 3-σ confidence

**UMBRA Singularity Rocket Team** — Avionics & Recovery Engineer, Systems Liaison *Cal Poly Pomona · Aug 2021 – May 2023*

- Integrated the embedded tracking and data-logging system for controlled maneuvers; 1st place at FAR 51025 with a 12,902 ft apogee and a $6,000 Lockheed Martin sponsorship

**CPP Liquid Rocket Lab** — Propellant Feed Systems Team (Bronco 1) *Cal Poly Pomona · Aug 2022 – May 2023*

- Contributed to feed-system documentation, component traceability, and failure analysis for a student liquid rocket's cryogenic LOX feed system; documented design changes and supported test readiness